EC39001: Analog Communication Lab
Experiment 2: Switching Modulator

Jai Raj
23EC10031
Group-6

Department of E&ECE
IIT Kharagpur

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1 Introduction

The objective of this experiment is to study the working of a switching modulator using a high-speed electronic switch, typically a transistor. The modulator responds to a carrier signal by turning the switch ON and OFF, thereby multiplying the carrier and the message signal.

2 Key Objectives

1. To implement a switching modulator using a transistor.
2. To apply a message signal and a carrier signal to the circuit.
3. To observe the output waveform of the switching modulator.
4. To design and use a Band-Pass Filter (BPF) for extracting the modulated signal.
5. To determine the following parameters using the transfer function of the BPF:
    - Mid frequency (f_m)
    - Gain at f_m
    - Quality factor (Q)
    - Bandwidth (B)

3 Components Used

- J-FET - BFW11 (high-speed switching type)
- Op -Amp - UA 741

- Resistors and capacitors (as per design)
- Function Generator
- Digital Storage Oscilloscope (DSO)
- Breadboard and connecting wires
- DC Power Supply

4 Theory

A switching modulator uses a transistor (or other high-speed electronic switches) to modulate a carrier signal with a message signal. The carrier, typically a square wave, controls the switch, allowing or blocking the message signal periodically. When the switch is ON, the message signal is passed; when it is OFF, the message signal is blocked. This produces a waveform equivalent to the multiplication of the carrier and message signal.

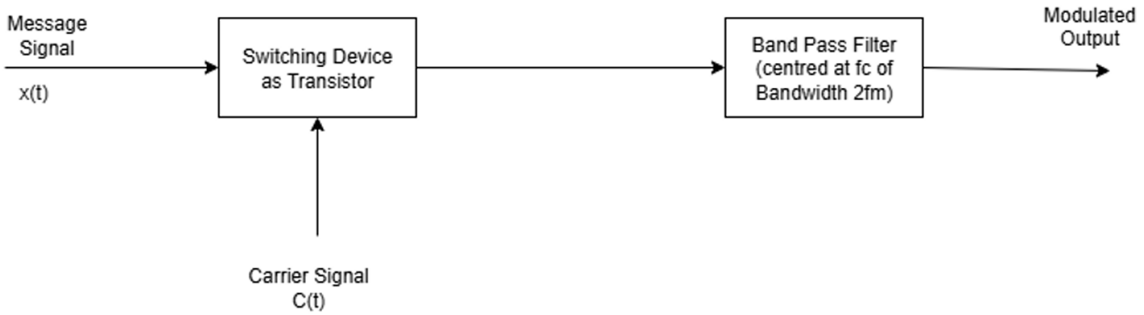


Figure 1: Block Diagram of Switching Modulator using a Transistor

Given Signals

- Message signal: $x(t)$ = sine wave, 4 Vpp, $f_m = 1$ kHz
- Carrier signal: $c(t)$ = square wave, duty cycle = 50%, $f_c = 10$ kHz

A Band-Pass Filter (BPF) is used to select the desired sideband from the modulated output. A second-order multiple-feedback band-pass filter is an active filter that uses an op-amp along with resistors and capacitors arranged in specific feedback paths. By combining the effects of low-pass and high-pass filtering, it allows a chosen range of frequencies around a specified center frequency to pass while attenuating others. The center frequency, bandwidth, and quality factor (Q) depend on the resistor and capacitor values. Owing to its high selectivity and sharp frequency response, this filter is widely used in communication systems to isolate desired signals.

The filter parameters can be determined from its transfer function:

- Transfer Funtion $A(s) = \frac{-\frac{R_2 R_3}{R_1+R_3} C\omega_m s}{1+\frac{2R_1R_3}{R_1+R_3} C\omega_m s+\frac{R_1R_2R_3}{R_1+R_3} C^2\omega_m^2 s^2}$

- Mid frequency $f_m = \frac{1}{2\pi C}\sqrt{\frac{R_1+R_2}{R_1 R_2 R_3}}$
- Gain at $f_m = -A_m = \frac{R_2}{2R_1}$
- Filter quality: $Q = \pi f_m R_2 C$
- Bandwidth: $B = \frac{1}{\pi R_2 C}$

A second-order multiple feedback type BPF is used for better selectivity.

5 Circuit Diagram

5.1 Switching Modulator

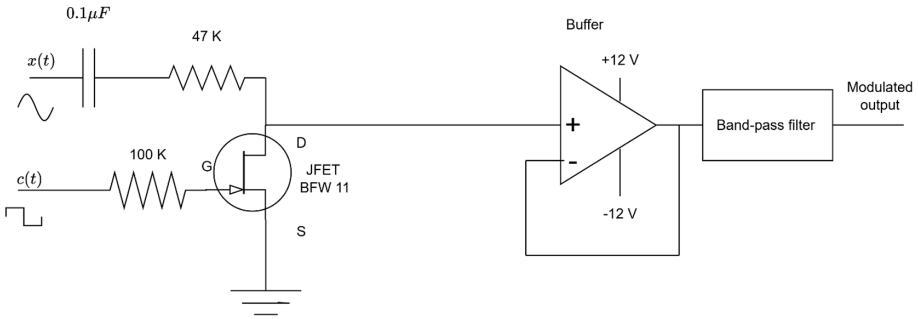


Figure 2: Switching Modulator using a Transistor

5.2 Band-Pass Filter

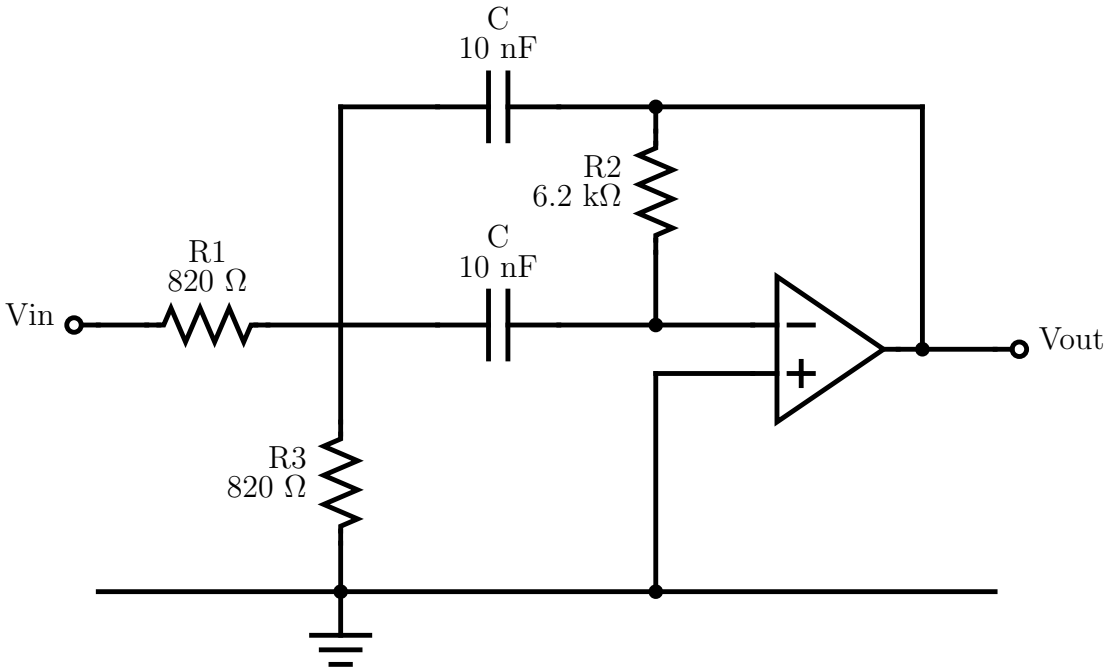


Figure 3: Second-order multiple feedback type BPF

- Components Value:
  - $R_1 = 820\,\Omega$
  - $R_2 = 6.2\ k\Omega$
  - $R_3 = 820\,\Omega$
  - $C = 10\,nF$

6 Expected Output

- The output of the switching modulator should show amplitude modulation, with the envelope corresponding to the message signal.
- The Band-Pass Filter output should contain the desired frequency component, with suppression of unwanted harmonics.
- The calculated f_m, gain at f_m, Q, and B should match theoretical values within acceptable tolerance limits.

7 Results and Discussion

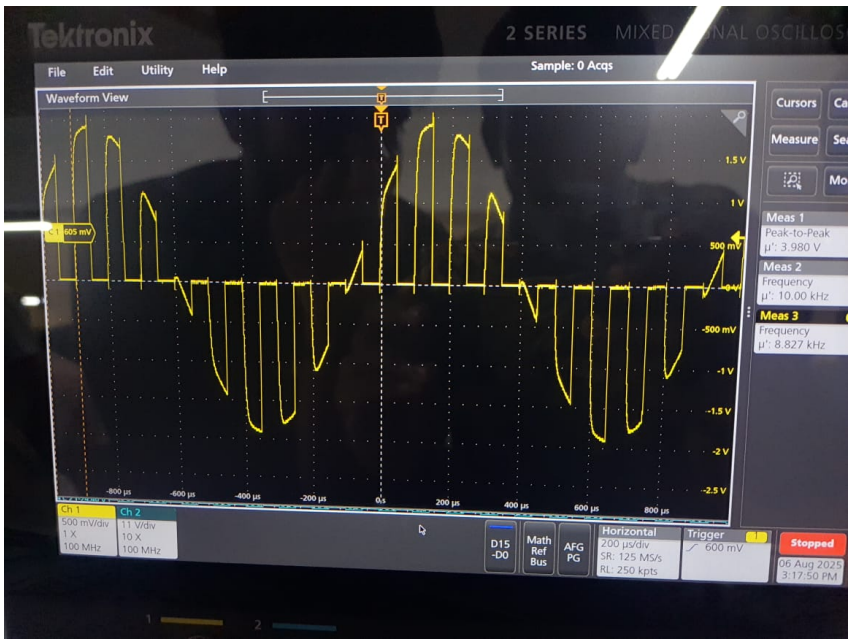


Figure 4: JFET Output

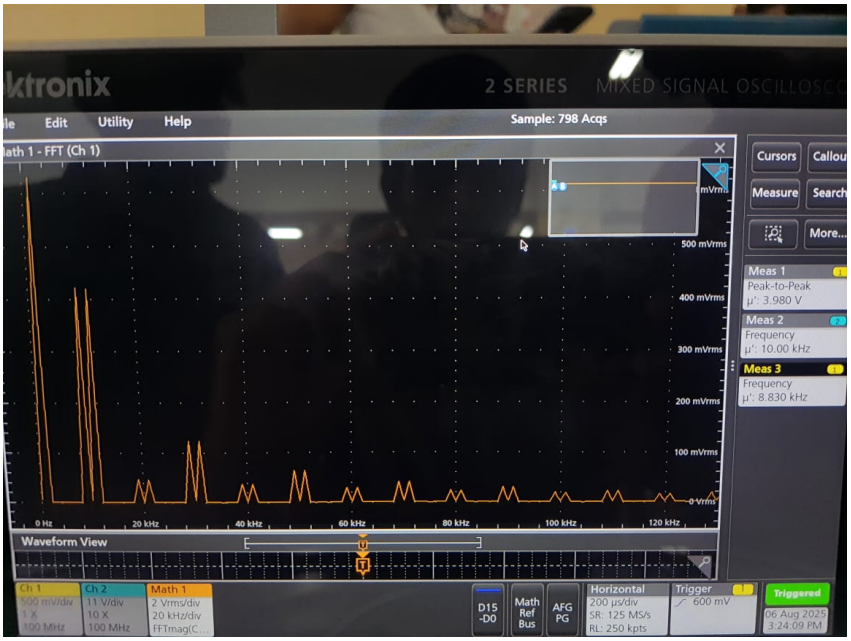


Figure 5: FFT of JFET Output



Figure 6: Buffer's Output

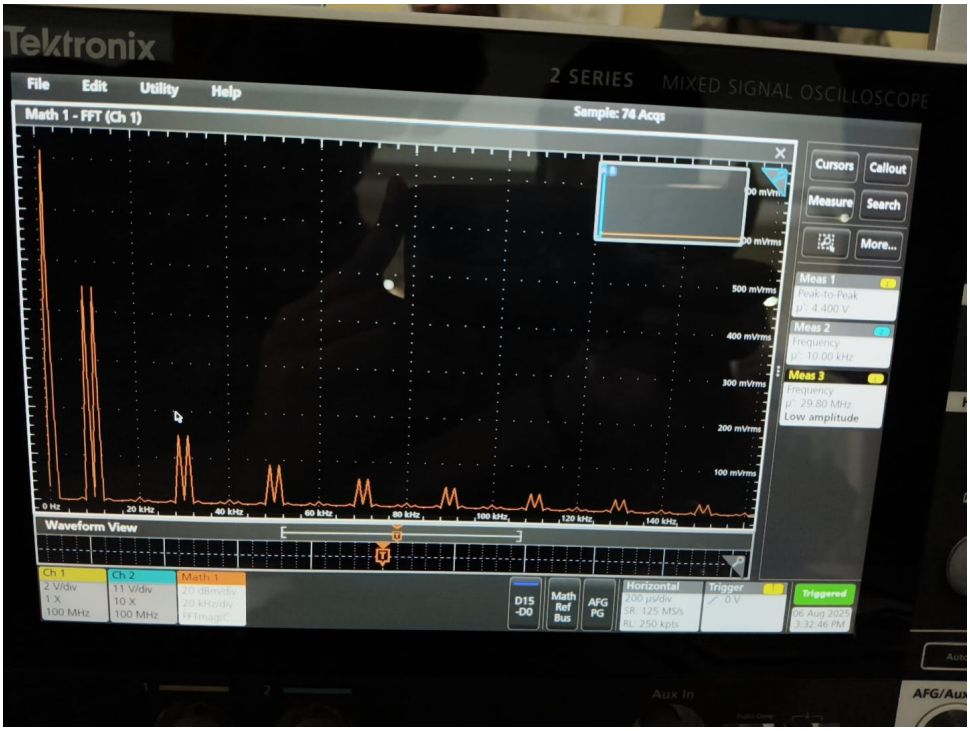


Figure 7: Buffer's Output FFT

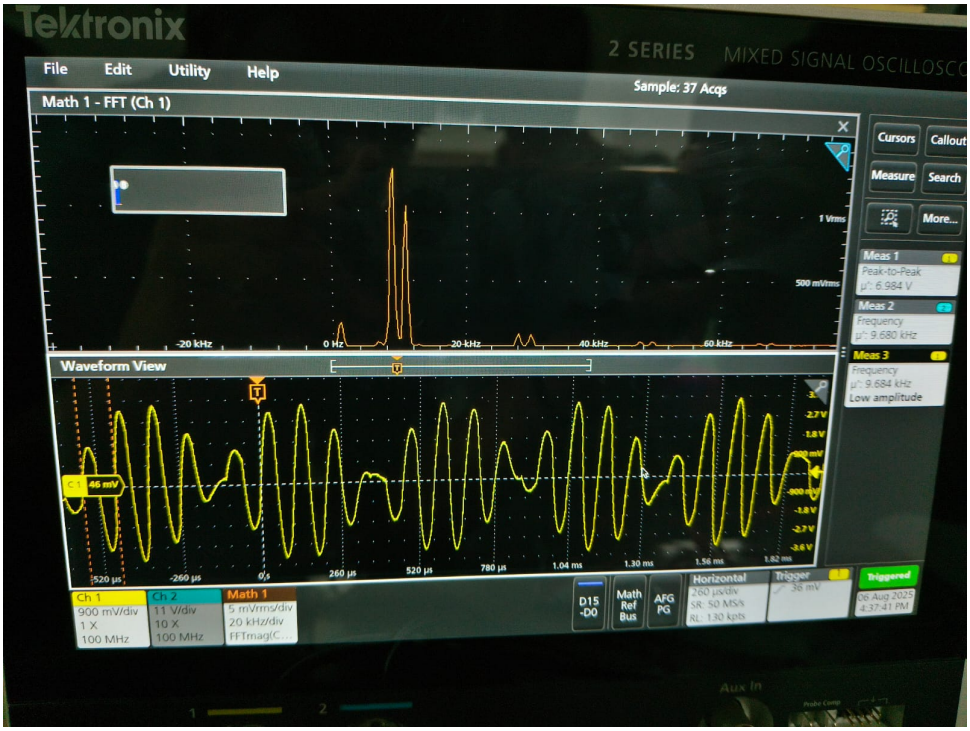


Figure 8: Modulated Siganl with FFT

A JFET-based switching modulator was built to modulate $x(t)$ with $c(t)$, using a DC offset of -2.5 V to bias the JFET in its switching region. This enabled amplitude modulation without a traditional multiplier. A buffer stage prevented loading effects, and a second-order BPF provided better selectivity and sharper roll-off than a first-order design.

A small mismatch between the BPF center frequency (10 kHz) and the actual signal (9.9 kHz) caused unequal FFT peak heights and slight spectrum asymmetry. This was due to approximate component values and non-ideal JFET switching. The modulated waveform was biased above the time axis to avoid phase reversals at zero crossings, at the cost of extra power consumption ($A^2/2$).

- **FFT Peak Height Variation:** In the final FFT , height of both the peaks are different(expected to be equal) because the BPF center frequency (10 kHz) is slightly higher than the actual signal (9.9 kHz), causing different Amplification at different points.
- **Presence of Sidebands:** The FFT shows sidebands around the carrier, as expected from amplitude modulation, with their relative heights matching theory.
- **Phase Shift in Output:** Sudden phase jumps occur when the carrier polarity flips from positive to negative and vice versa, inherent to switching modulation.
- **Effect of Buffer Stage:** The buffer slightly increases signal stability and amplitude but does not remove switching-induced phase shifts.
- **Spectral Purity Improvement:** After filtering, most high-frequency harmonics are attenuated, leaving a cleaner spectrum.
- **Component Tolerance Impact:** Small deviations in measured f_m, Q, and gain are due to resistor/capacitor tolerances.
- **Oscilloscope FFT Resolution:** Limited FFT bin resolution slightly affects the exact peak frequency and amplitude measurement.
- **Modulation Depth Observation:** Time-domain plots show clear envelope variation, indicating correct multiplication of carrier and message.
- **Overall Performance:** Results match theoretical predictions, confirming correct operation of the switching modulator and BPF.
- Measured values:
  - Mid frequency f_m=9.98 kHz
  - Gain at f_m: $A_m = 7.56$
  - Quality factor (Q)= 0.257
  - Bandwidth (B): 5.134 kHz
- Observations were consistent with the theory, with small deviations due to component tolerances and signal distortion in the switching process.

——END OF REPORT——